



Food Plant Maintenance Benchmark Pack

How food and beverage manufacturers structure maintenance — what they run, what they replaced, and a one-page readiness check for your own plant.

Get in touch

Aarti More
CARL Software Executive
iEngineering Australia
Email: aarti@iengaust.com.au
Phone: 02 5017 7866

INTRODUCTION

Most food plants run maintenance on three things

Aspreadsheet,awhiteboard,and oneperson'smemory. It works— rightupuntilabearing goes on the packing line at 2am, the spare is not in stock, and a shift of chilled product goes in the bin.

This pack is not a software brochure. It is a comparison — how food and beverage manufacturers have structured their maintenance function, what each of them replaced, and how large the operation was when they did it.

We published it because the question plant managers in this region ask most is not *"what does a CMMS do?"* It is *"what do plants like mine actually run, and when did they change?"*

What food manufacturing does to maintenance

Failure is not downtime. It is scrap.

On fresh and chilled lines, an unplanned stop does not only cost production hours — it can write off the batch in the machine and the batch behind it.

The plant is washed down every day

Daily cleaning and disinfection is hard on bearings, seals and sensors in a way that general manufacturing never sees.

Audits ask for proof, not intent

BRCGS and FSSC 22000 expect documented, scheduled maintenance of food-contact equipment and clean release back to production.

Spares can be months away

One chocolate manufacturer in this pack reports lead times of four to six months on some parts — a stock policy problem long before it is a breakdown problem.

How to use it

- **Pages 3–4** — six practices that separate plants in control of maintenance from plants reacting to it.
- **Pages 5–7** — four manufacturers in detail: what changed, and what it took.
- **Pages 8–9** — a benchmark table of deployments by sector and scale.
- **Pages 10–12** — a readiness checklist you can score in ten minutes, whatever system you choose.

BENCHMARK · PART ONE

Six practices that show up in every plant that has this under control

None of these require a particular product. They are the operating habits the plants in this pack have in common, and they are what any system you buy has to support.

1 Preventive work is scheduled per machine, and it actually fires

Not a spreadsheet of intentions. Each asset carries its own schedule, built from the manufacturer's specification and adjusted by what the equipment specialists observe. Plants that do this well track the ratio of preventive to corrective work as a management number, and defend it.

In one multi-country malt producer, structuring maintenance this way took the preventive maintenance rate to 30 per cent of all work.

2 Spares and stores are tied to the work order, not to a storeman

The parts needed for a job are reserved when the job is planned. Minimum levels are set against real consumption. One fresh-pasta producer integrated the system with its automated storage towers so parts are picked as the job is prepared, rather than looked for when the technician arrives.

3 Technicians report from the floor, on a phone, not on paper

Mobile access to equipment history, technical specifications and breakdown records at the machine — and the report closed in a few taps before the technician walks away. This is the single change that most improves the quality of maintenance history, because it removes the gap between doing the work and writing it down.

BENCHMARK · PART TWO**4 Operators can raise a fault without a licence or a phone call**

The people who notice a problem first are rarely maintenance staff. In the larger deployments in this pack, the number of occasional requesters dwarfs the number of maintenance users — hundreds or thousands of operators with a simple request screen, feeding a handful of planners. Requests are then tracked from issue to closure, so nothing is lost between a shift and a supervisor.

5 Contractors and regulatory work sit in the same system as everything else

Subcontracted work, statutory inspections and calibration are where most plants lose visibility, because that work lives in a different folder and a different inbox. Plants with this under control run contractor access into the same system, against the same asset register, so a compliance obligation is a scheduled work order like any other.

6 Equipment history is complete enough to hand to an auditor

The test is simple. Pick a line. Pick a month eighteen months ago. Ask who maintained it, when, against what specification, and what was replaced. If that takes an afternoon of phone calls, the history is not a record — and it will leave the business the day the person holding it does.

The pattern behind all six

Every plant in this pack changed systems for the same underlying reason: not because the old one broke, but because the operation outgrew it.

A second site

A tool chosen for one plant rarely survives the second one — reporting, spares and standards diverge immediately.

Contractors

The moment external labour carries regulatory work, a spreadsheet stops being defensible.

A doubled plant

Capacity expansion is the most

common trigger of all — the asset count passes what the old method can hold.

Read the next three pages with that in mind. The useful question is not whether these companies are like yours in size, but whether the trigger that moved them is one you can already see coming.

CASE STUDY ONE

Malteurop — one method across fourteen countries

One of the world's largest malt producers, operating in 14 countries including Australia, the United States, France and Germany, with 10,500 partner grain growers. It is estimated that one in ten beers is brewed with its malt.

**10,000**

equipment units under management

12,000

service calls tracked per year

200+

daily users, multiple languages and currencies

The trigger

In 2016 the group launched a maintenance modernisation plan. The objective was not to fix a broken system — it was to make fourteen countries work the same way: standardise maintenance practice, improve the safety of maintenance and production teams, reduce corrective work, improve machine cost prediction, and connect maintenance to the SAP ERP so orders and budget could be tracked in one place.

What they did

Deployment began in 2017 with Franco-German project teams and workshops with technicians every fortnight. Existing data was mass-imported rather than re-keyed. The desktop system was paired with a mobile application so technicians could reach technical specifications and breakdown history on their phones and file reports in a few clicks.

Where it landed

More than 200 people now use the system daily. Inventory covers over 10,000 items. The preventive maintenance rate stands at 30 per cent, and performance indicators the group designed itself are used to analyse all 12,000 annual service calls. The rollout has continued into Australia and the United States.



“CARL Source Factory gives us excellent data visibility and accessibility.”

Xavier Niquet, Group Maintenance Manager, Malteurop

Why it matters here: this is not a European reference. Malteurop already runs this system in Australia — the same platform, supported in this region by iEngineering.

CASE STUDY TWO

Saint Jean — when the plant doubled, the system could not follow

A regional fresh pasta and ravioli producer of about 520 people, running four plants. The main site at Romans-sur-Isère employs roughly 380 and runs fourteen production lines, from raw material and dough through rolling mills, calibrators, forming, thermoforming, packaging and palletising.

20,000 → 40,000
m²

expansion of the main plant that
triggered the change

14

production lines at the main site

25

people in the site maintenance
department

The trigger

Demand rose and the main plant doubled in size. The previous system no longer met the company's scalability needs, and a new one was implemented as part of the upgrade, deployed across 2016 and 2017.

What makes this plant instructive

It is a mixed estate — highly automated lines such as sheet grippers for packaging sit alongside predominantly mechanical machines like rolling mills and calibrators, washed and disinfected every day. The maintenance department is organised accordingly: equipment specialists monitoring recurring breakdowns, a methods team scheduling preventive work against manufacturer specifications, a team on spares and purchasing, and automation and energy specialists working across all sites.

Where it landed

The system is integrated with the site's automated storage towers, so parts are prepared with the job. It manages procurement and tracks contracts for subcontracted work, particularly regulatory compliance. Management monitors line health through preventive and corrective ratios per site — including to justify additional maintenance resource. With connected machines, the predictive module flags faults such as a failing variable speed drive or a clogging filter, and proposes the action.

“We operate on a just-in-time basis because of the fresh nature of our products. We cannot afford for a breakdown to disrupt an entire workshop.”

Stéphane Gerbert, CMMS Referent and Infrastructure Expert, Saint Jean

CASE STUDY THREE

Grain de Sail — the smallest plant in this pack

A coffee roaster and chocolate maker that grew from a workshop into an industrial manufacturer, producing 500 tonnes of chocolate a year from a 2,500 m² factory, with cocoa carried across the Atlantic by sailing cargo ship.

50+

machines in one chocolate factory

500 t

chocolate produced per year

4–6 months

reported lead time on some spare parts

What makes this case useful is the timing. The specification was written in 2020 **while the plant was still being built** — multi-site management, SaaS operation and the intended workflows were decided before the first machine ran. Preventive maintenance was scheduled from the outset rather than retrofitted after the tenth breakdown.

The technical manager is candid that a new plant is not an automatic success: the team still had to learn the machines. Production and maintenance built their own screen forms, which ended the reliance on oral history, and preventive routines were tuned as wear patterns emerged — for example on the conche, whose blades wear against the vat walls.

“Beyond 3 or 4 machines, CMMS is essential. No argument can justify doing without.”

Jean-Marc Riou, Industrial Technical Manager, Grain de Sail

CASE STUDY FOUR

Lindt — the same platform at group scale

At the other end of the range, the Swiss chocolate manufacturer runs the same platform across production sites in six countries, with roughly 300 desktop users and 200 mobile users, interfaced with SAP and operating in more than one language.

Between these two cases sits the practical argument for choosing carefully once: a single factory with fifty machines and a multinational group with six production countries are running the same system. Nothing has to be replaced on the way up.

BENCHMARK

Twenty food and beverage deployments

Sector, scale and the shape of the user base. Where a company is not named, it is described by sector only. “Desktop” covers planners, engineers, stores and purchasing; “mobile” covers technicians on the floor; “requesters” covers operators who only raise faults.

Sector	Scale	User base	Notable
Malteurop — malt	14 countries	200+ desktop & mobile	10,000 equipment units; SAP; 30% preventive
Lindt — chocolate	6 countries	~300 desktop, ~200 mobile	SAP interface; multi-language
Saint Jean — fresh pasta	4 plants	10 desktop, 7 mobile	Automated store integration; predictive
Grain de Sail — chocolate	1 factory	Small core team	Specified before the plant was built
Meat and poultry processing	31 plants	170 desktop, 275 mobile, 1,000 requesters	1,445 users in total
Multi-site food group	62 production units	90 desktop, 40 mobile	SAP interface at each equipped site
Grain trade cooperative	~20 companies	11 desktop, 60 mobile	12,000 maintenance requests a year
Biscuits and bakery	8 plants	120 desktop, 4 mobile	Interfaced to time and finance systems
Chocolate and confectionery	Multi-plant	20 desktop, 11 mobile, 250 requesters	281 users in total
Spirits bottling	3 sites	60 desktop	40,000 bottles per hour

The four named deployments above are covered in detail on pages 5 to 7.

BENCHMARK

Twenty food and beverage deployments

Continued. Named entries are companies with published success stories; the remainder are described by sector and scale only.

Sector	Scale	User base	Notable
Fruit and vegetable cooperative	1,000 growers	2 desktop, 6 mobile, 50,000 requesters	Contractor access included
Pet food manufacturing	45,000 m ²	10 desktop, 10 mobile, 600 requesters	Runs 24 hours, 360 days a year
Flexible packaging	4 factories	3 desktop, 250 requesters	Packaging for food manufacturers
Pig processing	16,000 m ²	5 desktop, 15 mobile, 15 contractor	10,000 head per day
Dairy cooperative	3,500 members	10 desktop, 10 mobile, 10 contractor	Multi-organisation configuration
Chilled and frozen food logistics	10,000+ m ² pilot	15 desktop, 15 mobile, 15 contractor	Fridge warehouse maintenance
Fine wines	Multi-site	4 desktop, 6 mobile	Equipment, work and stock modules
Vegetable seed production	2 sites	5 desktop, 5 mobile	Research and farming facilities
Collective catering	Multi-kitchen	2 desktop, 2 mobile, 2 contractor	Kitchen equipment and interventions
Cocoa processing	Multi-country	Pilot then extension	Replaced an incumbent enterprise system

Two things to take from this table. First, the range: the same platform is running a two-person kitchen operation and a 1,445-user meat group. Second, the shape: in every large deployment the requester population is far bigger than the maintenance population — which is how faults get reported at all.

READINESS CHECKLIST

Score your own plant in ten minutes

Answer honestly about how the plant runs today, not how it is meant to run. Tick Yes only where it is true for every production line, not just the newest one. Scoring follows on page 12.

A THE ASSET REGISTER

- | | | |
|---|--------------------------|--------------------------|
| 1. Every production asset has a unique identifier used by both maintenance and production. | ☐ | ☐ |
| 2. You can list every asset on a given line without asking a specific person. | ☐ | ☐ |
| 3. Assets are grouped by operating system (intake, packing, utilities), not only by location. | ☐ | ☐ |
| 4. Critical spares are linked to the assets that use them. | ☐ | ☐ |

B PREVENTIVE MAINTENANCE

- | | | |
|---|--------------------------|--------------------------|
| 5. Every critical asset has a written preventive schedule. | ☐ | ☐ |
| 6. Those schedules are derived from manufacturer specifications, not habit. | ☐ | ☐ |
| 7. You know what proportion of last month's work was preventive rather than corrective. | ☐ | ☐ |
| 8. Preventive work that was skipped last month is visible to a manager. | ☐ | ☐ |

C WORK CAPTURE

- | | | |
|--|--------------------------|--------------------------|
| 9. An operator can report a fault without finding a supervisor. | ☐ | ☐ |
| 10. Technicians record what they did at the machine, the same day. | ☐ | ☐ |
| 11. A request cannot be closed without a record of what was actually done. | ☐ | ☐ |
| 12. You can see how long faults waited before someone started on them. | ☐ | ☐ |

Left column: Yes · Right column: No

READINESS CHECKLIST

Score your own plant in ten minutes

Continued — sections D and E.

D SPARES AND PROCUREMENT

13. Minimum stock levels are set from consumption, and reviewed.	☐	☐
14. Long lead-time parts are identified and held deliberately.	☐	☐
15. Parts are reserved when a job is planned, not hunted when it starts.	☐	☐
16. You know what maintenance cost you last year, by line.	☐	☐

E COMPLIANCE AND CONTINUITY

17. Statutory inspections and calibrations are scheduled in the same place as other work.	☐	☐
18. Contractor work is recorded against the asset, not only in an invoice.	☐	☐
19. You could produce the maintenance record for one line, for one month last year, within an hour.	☐	☐
20. If your most experienced maintenance person left tomorrow, the plant would still know how it is maintained.	☐	☐

Left column: Yes · Right column: No

READINESS CHECKLIST

What your score means

Count the Yes answers. The bands below are not a sales ladder — they describe what usually needs attention first.

0–6 Reactive

Maintenance is held together by individuals and memory. The risk is concentration: one resignation and the plant loses how it is maintained. The first job is not software — it is an asset register and a way for operators to report faults. Both can be done in weeks.

7–12 Organised but fragile

Preventive routines exist and someone is keeping them. They usually survive normal weeks and collapse in busy ones, and nobody can prove afterwards what was skipped. Focus on capture at the machine and on making skipped preventive work visible to a manager.

13–17 In control, not yet in evidence

The work is being done well. The gap is proof and cost — audit-ready history, contractor work recorded against assets, and maintenance cost by line. This is the band where a proper system pays for itself fastest, because the practices already exist and only need a home.

18–20 Ready to optimise

You are in the territory where condition monitoring and predictive maintenance return something. The next gains come from connected machines, vibration and ultrasound on slower equipment, and linking maintenance data to overall equipment effectiveness.

Three questions worth more than the score

- **Question 19** — if one line-month of history takes longer than an hour to produce, you have an archive, not a record.
- **Question 20** — plants that answer No here, and later lose that person, all describe the same eighteen months of rebuilding.
- **Question 7** — a plant that cannot state its preventive ratio cannot tell whether anything it changes is working.

Want a second opinion on your score? Send us an asset list — the 30-minute review on the last page.

REFERENCE

What the plants in this pack are running

CARLSource Factory, from CARL Berger-Levrault— an enterprise asset management and maintenance platform used across industry, utilities, transport, healthcare and government. CARL Source Facility covers buildings and multi-site property. Both run as SaaS or on-premise, and are multi-site, multi-language and multi-currency.

Access follows how a plant actually works

Module	Who uses it	What it is for
CARL Source	Planners, engineers, stores, purchasing	The full system — asset register, work orders, preventive schedules, stock, procurement, contracts, reporting
CARL Touch	Technicians on the floor	Mobile access to specifications and history at the machine; reports closed in a few taps
CARL Flash	Operators and occasional users	A simple fault and request screen that scales to thousands of people without full licences
CARL Xpress	External contractors	Contractor access to their own work against your asset register
CARL Maps	Utilities, networks, campuses	Linear and mapped assets — water, rail, road, distribution networks
BL.Predict	Reliability teams	Predictive maintenance from connected machine data
CARL BI Analyses	Managers	Activity reporting that non-specialists can read

Integration

The deployments in this pack interface with SAP and other ERP systems, MES, automated storage, time and attendance and finance systems. A REST API and standard web services are available for anything else. Existing data is mass-imported at go-live rather than re-keyed.

Where iEngineering fits

We are an engineering company, not a software reseller. iEngineering works across Australia, New Zealand, India, South East Asia and the Pacific, and we implement, configure, train and support CARL ourselves — in your time zone, on your site, by engineers who have stood in plant rooms. That is the part of this that does not appear in a feature list and matters most at 2am.

NEXT STEP

Send us your asset list

The most useful half hour you can spend after reading this is not a demonstration. It is a look at your own equipment.

The 30-minute asset register review

Send us an equipment list for one line or one site — a spreadsheet is fine, however rough.

We will come back with a preventive maintenance structure for it: how the assets group, where the schedules should sit, which spares are worth holding, and what your readiness score means in practice.

No charge, no obligation, and the structure is yours whatever you decide to do next.

Talk to us

enquiries@iengaust.com.au

+61 (02) 8318 1490

ieng.tech

Where we work

Australia and New Zealand · India · South East Asia ·
Fiji, Papua New Guinea and the Pacific islands ·
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Also from iEngineering

Power system studies, ASP Level 3 design, substation and industrial design, earthing and lightning analysis, renewable and battery energy systems, industrial automation, and electrical and energy audits.

Other engineering software

We are also the Asia-Pacific partner for NEPLAN power system analysis and XGSLab earthing, EMFI and lightning analysis.

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